#vlsi #characterization #TEM #electron_microscope

transmission electron microscope

a high energy electron beam goes **through** a very thin sample, and you image what comes out the other side. it's the closest thing to a light microscope but with electrons, see Electron microscope. part of Metrology and nano-characterization workshop

- typical 80–300 kV → $\lambda \approx 2$–4 pm
- resolution <0.1 nm → you can see **individual atomic columns**
- gives a 2D **projection** of the inside, not the surface

> **THE ONE IDEA**
>
> a TEM has a real lens after the sample, so it can show you either the image of the sample or its diffraction pattern (its Fourier transform), just by changing one lens. that's what makes it both a microscope and a mini XRD

the wavelength

the same de Broglie wavelength as in the SEM

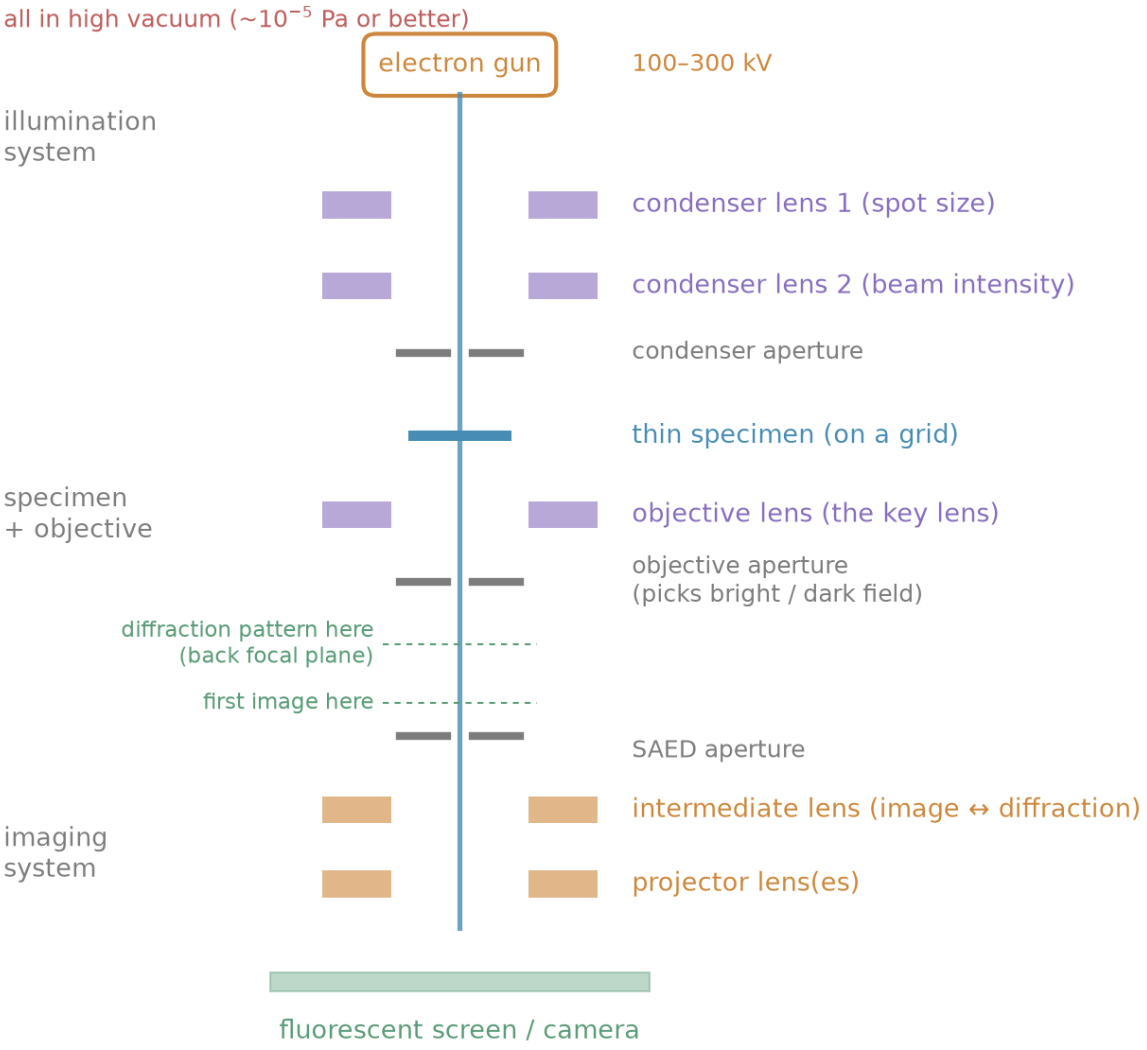
$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2m_e eV}} \approx \frac{1.226}{\sqrt{V}}\ \text{nm} \qquad (V \text{ in volts})$$

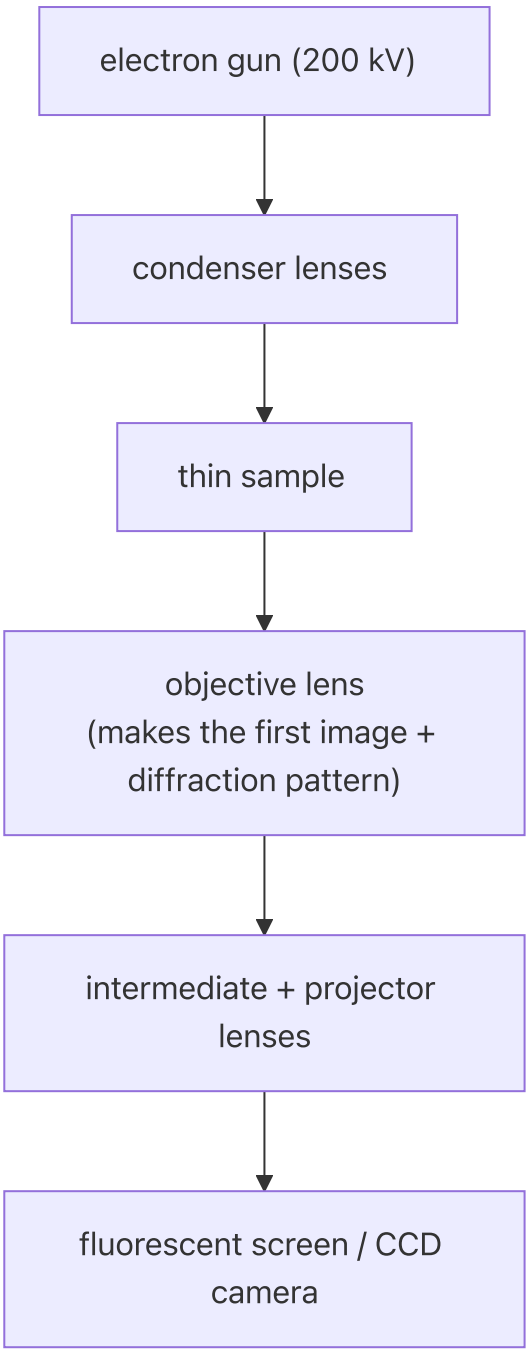
at TEM voltages the electrons move at more than half the speed of light, so you need the **relativistic** version (the extra bracket makes λ a bit shorter)

$$\lambda \approx \frac{1.226}{\sqrt{V\left(1 + 0.9785 \times 10^{-6}\, V\right)}}\ \text{nm}$$

> **THE WAVELENGTH AT 200 KV**
>
> $$V\left(1 + 0.9785 \times 10^{-6}\, V\right) = 200\,000 \times (1 + 0.1957) = 239\,140$$
> $$\sqrt{239\,140} = 489.0$$
> $$\lambda \approx \frac{1.226}{489.0} = 0.00251\ \text{nm} = 2.51\ \text{pm}$$
>
> without the correction you'd get $\frac{1.226}{\sqrt{200\,000}} = 2.74$ pm, about 9% too long. and it's ~60× shorter than the X-rays in XRD (154 pm)

parts


all in high vacuum (~10⁻⁵ Pa or better)
electron gun
100–300 kV
illumination system
condenser lens 1 (spot size)
condenser lens 2 (beam intensity)
condenser aperture
thin specimen (on a grid)
specimen + objective
objective lens (the key lens)
objective aperture (picks bright / dark field)
diffraction pattern here (back focal plane)
first image here
SAED aperture
intermediate lens (image ↔ diffraction)
imaging system
projector lens(es)
fluorescent screen / camera



the **intermediate lens** decides what you see, switch it one way and you see the **image**, another way and you see the **diffraction pattern**

electron sources

field emission is ~1000× brighter with a 10× smaller energy spread than tungsten, but needs a far better vacuum

	Tungsten	LaB_6	Field emission
work function (eV)	4.5	2.4	4.5
operating temperature (K)	2700	1700	300
current density (A/m^2)	5×10^4	10^6	10^{10}
crossover size	50 µm	10 µm	<0.01 µm
brightness ($A/m^2/sr$)	10^9	5×10^{10}	10^{13}
energy spread (eV)	3	1.5	0.3
emission current stability (%/hr)	<1	<1	5
vacuum (Pa)	10^{-2}	10^{-4}	10^{-8}
lifetime (hr)	100	500	>1000

characteristics of the three principal electron sources operating at 100 kV (values from the lecture slide)
field emission: ~1000× brighter, 10× smaller energy spread, but needs a far better vacuum

(thermionic = boil electrons off a hot filament. field emission = pull them out with a huge electric field, which is really **quantum tunnelling**, see the quantum connection)

sample has to be thin

method	good for	watch out for
FIB lamella	exact site, chips, devices	Ga damage layer
ultramicrotome	soft / polymer / biological	knife marks
ion milling (Ar^+)	ceramics, large area	no site selection
drop cast on grid	nanoparticles, powders	stacking / clumping

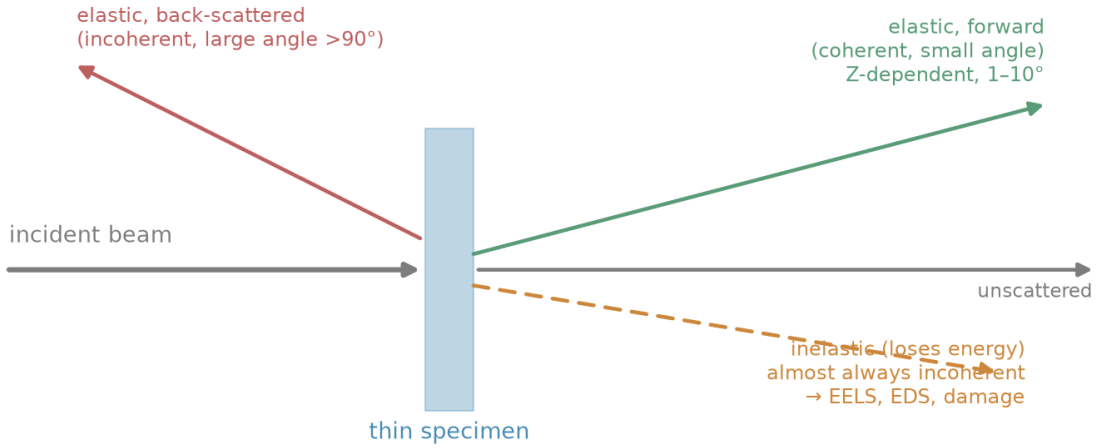
all of them aim for a specimen under ~100 nm thick (<50 nm for atomic resolution)

- thickness **<100 nm** (for atomic resolution <~50 nm) or the electrons get absorbed or scatter multiple times
- how to make a thin enough slice
  - FIB lamella (the standard for chips, you pick the exact spot)
  - ultramicrotome (diamond knife, for soft stuff)
  - ion milling / dimple grinding
  - or just drop nanoparticles on a carbon coated copper grid

- sample prep is the hard and slow part of TEM, much harder than for an SEM

scattering inside the specimen

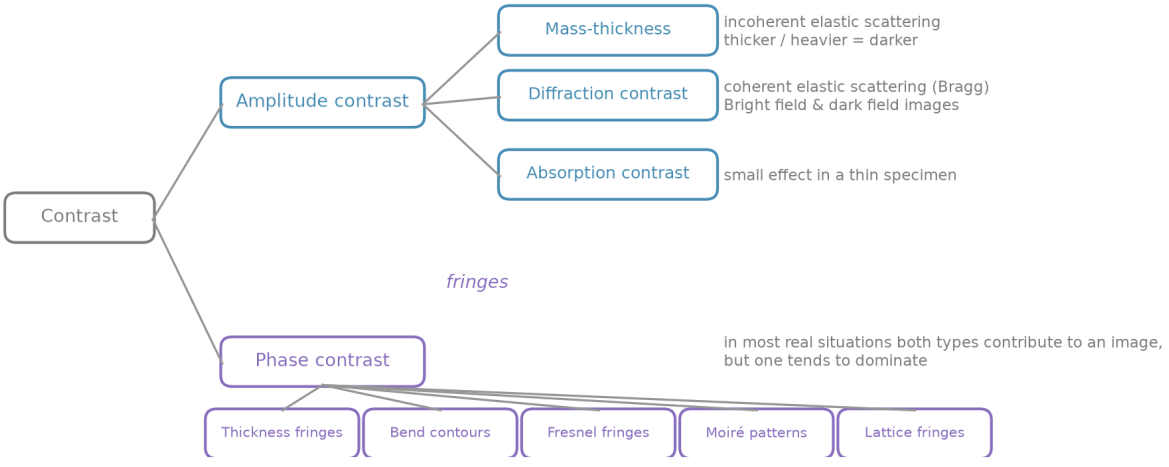
elastic scattering in a thin crystalline specimen is mostly coherent and forward (that's what makes diffraction), inelastic is almost always incoherent



thin + crystalline → mostly coherent forward scattering → diffraction. thick → mostly incoherent.

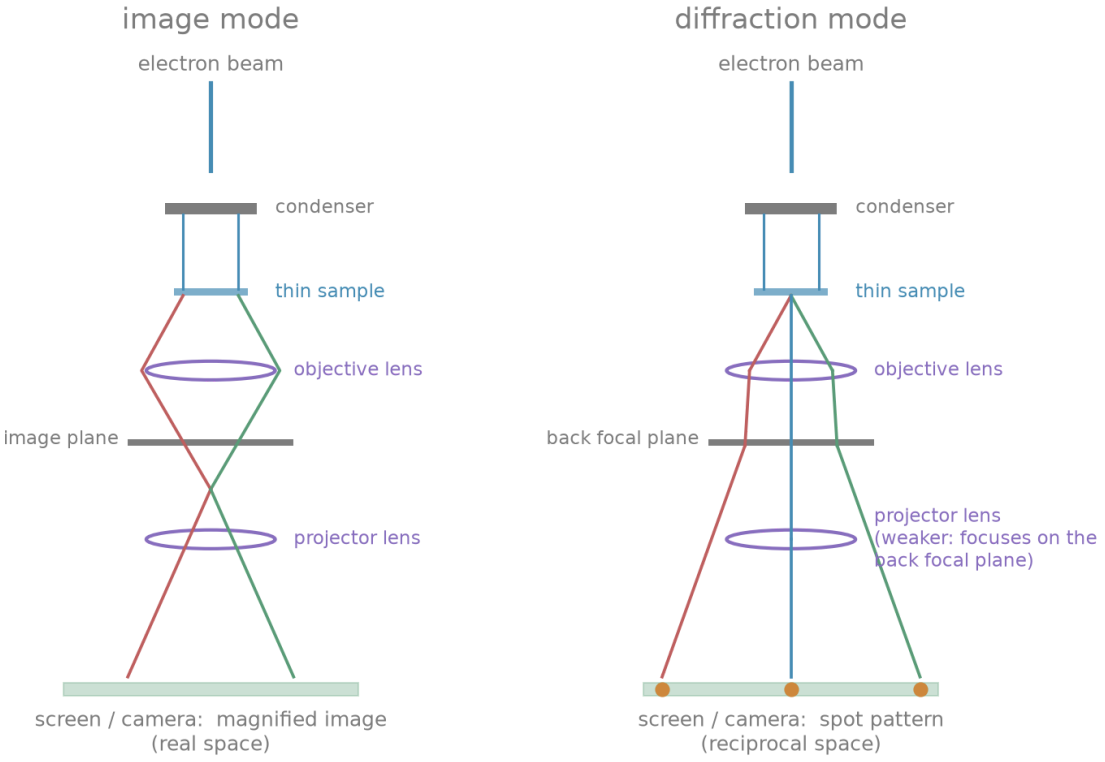
contrast, why do you see anything

- **mass-thickness contrast** → thick or heavy parts scatter more, so look darker
- **diffraction contrast** → crystals scatter strongly at certain angles (Bragg, see XRD > Bragg's law)
- **phase contrast** → **interference** between the electron waves, this is what gives the atomic lattice fringes in HRTEM



modes

mode	what you do	you get
bright field	aperture lets only the unscattered beam through	scattering regions dark
dark field	aperture picks a diffracted beam	only the grains satisfying that diffraction light up
SAED (selected area electron diffraction)	select an area, view the back focal plane	spot pattern → crystal structure and orientation (rings = polycrystal, halo = amorphous)
HRTEM	big aperture, phase contrast	lattice fringes, atomic arrangement, interfaces
STEM	scan a small probe through the sample	with HAADF detector, brightness $\sim Z^2$ so you get atomic number contrast
EDS / EELS in STEM	analyze X-rays / energy lost	element maps at the atomic scale



same sample, same beam. Only the strength of the intermediate / projector lens changes

(spots / rings / halo is the same single crystal / polycrystal / amorphous story as the XRD patterns. and EELS can measure a bandgap too, like UV-vis, but from a tiny spot)

Bragg's law in a TEM

the same law as in XRD

$$n\lambda = 2d\sin\theta$$

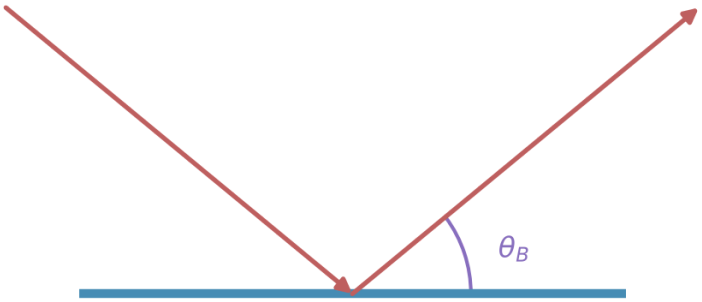
λ is so short that the Bragg angles are under a degree, so $\sin\theta \approx \theta$

THE SI (111) PLANES AT 200 KV

$d_{111} = \frac{5.431}{\sqrt{3}} = 3.136$ Å and $\lambda = 0.0251$ Å (from above)

$$\theta \approx \frac{\lambda}{2d} = \frac{0.0251}{2 \times 3.136} = 0.0040 \text{ rad} = 0.23^\circ$$

the same planes in XRD with Cu K_α diffract at $\theta = 14.2^\circ$. so in a TEM all the diffracted beams go almost straight on, and the pattern on the screen is a flat picture of the crystal's **reciprocal lattice**



(111) planes, d = 2.1 Å (Cu)

$$2d\sin\theta_B = n\lambda$$

120 kV electrons: $\lambda = 0.0335$ Å

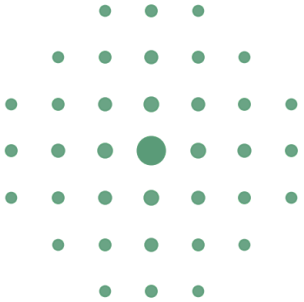
$$\theta = \sin^{-1}\frac{0.0335}{2\times 2.1} = 0.46^\circ$$

= 7.97 mrad (10 mrad = 0.573°)

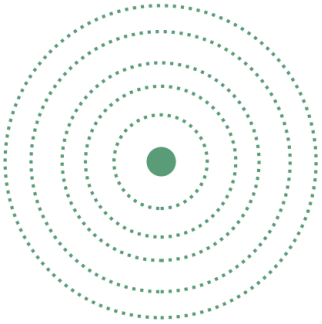
angles in a TEM are tiny, so you can use $\theta \approx \lambda/2d$

X-rays at 1.54 Å need θ ≈ 21° for the same planes, electrons need under 1°

single crystal: spots



polycrystal: rings



amorphous: diffuse halo



spot distance R from the centre → plane spacing d through $Rd = \lambda L$ (the camera equation)

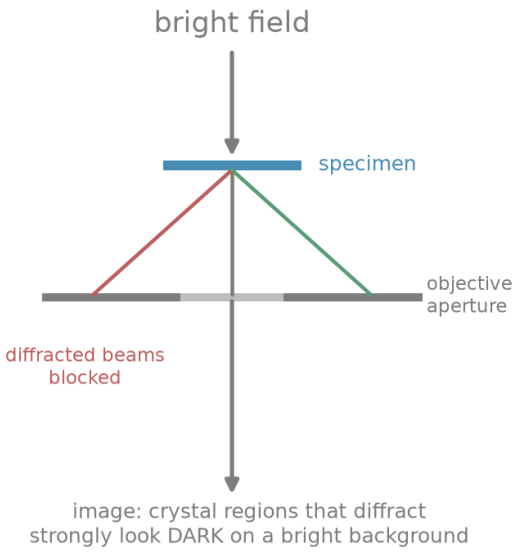


image: crystal regions that diffract strongly look DARK on a bright background

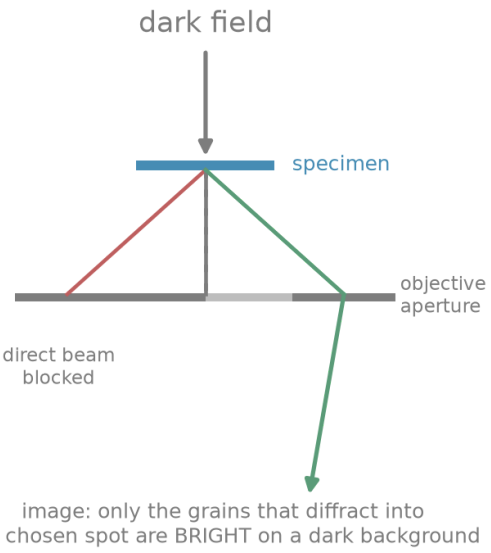
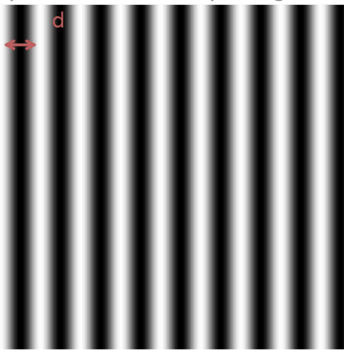
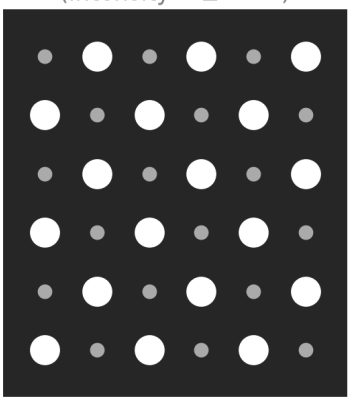


image: only the grains that diffract into the chosen spot are BRIGHT on a dark background

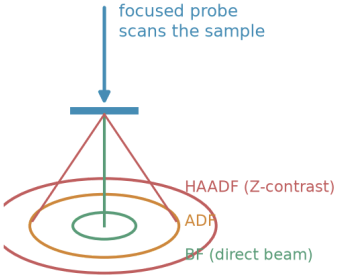
HRTEM: lattice fringes (phase contrast, spacing = d)



HAADF-STEM: bright = heavy atoms (intensity $\propto Z^{1.6-2}$)



STEM detectors



STEM: probe scanned, detectors made of rings at different angles

vs SEM

	SEM	TEM
electrons	scattered back / emitted	transmitted
sample	thick OK	<100 nm
resolution	~1 nm	<0.1 nm
tells you	surface	inside, crystal structure
effort	easy prep	hard prep, costly

in VLSI

- **gate stack** → measure gate oxide thickness (a few atoms of high-k)
- **fins / nanosheets** → exact shape and width
- **interfaces** → epitaxy quality, interlayer, defects, dislocations
- thin film layers in Interconnects (barrier, seed, Cu)
- failure analysis at the nm scale

BEAM DAMAGE

TRY THESE

① find λ at 300 kV

② a lamella is 250 nm thick. what goes wrong?

③ an SAED pattern shows sharp rings. what does the film look like?

④ which mode would you use to see if a 2 nm gate oxide is crystalline or amorphous?

ANSWER

the quantum connection

THE LENS DOES A FOURIER TRANSFORM

the diffraction pattern in the back focal plane (what you see in SAED mode) is the **Fourier transform** of the specimen, done by the objective lens itself. a crystal repeats, so the pattern is a grid of sharp spots, the same "repeating signals → sharp spikes" rule as in Fourier Transform > Why it matters for quantum and XRD > the quantum connection

switching between image and diffraction mode is switching between "real space" and "Fourier space", just like a Quantum Fourier Transform switches a qubit register between the computational basis and the Fourier basis

INTERFERENCE OF ONE ELECTRON AT A TIME

phase contrast (HRTEM) is **interference** of electron waves. the beam current is so low that usually there's only one electron in the column at a time, so each electron interferes **with itself**, the same as a single photon in a Mach–Zehnder interferometer

FIELD EMISSION IS QUANTUM TUNNELLING

a strong electric field makes the energy barrier at the tip so thin (~nm) that electrons **tunnel** straight through it, with no heating. a classical particle could never cross a barrier higher than its energy. the chance of getting through drops **exponentially** with the barrier's thickness, which is why the field has to be so strong

CHECKING QUBITS

the heart of a superconducting qubit is a **Josephson junction**: two superconductors separated by an oxide barrier only ~1–2 nm thick that electron pairs **tunnel** through. cross-section TEM (on a lamella cut by FIB) is how that barrier is measured. because tunnelling is exponential in the thickness, a tiny change in the barrier changes the junction a lot, and that sets the qubit's frequency

see also Electron microscope, FIB, SEM, EDS, XRD, UV-vis spectroscopy, Metrology and nano-characterization workshop

formulas: Microscopy formulas